

Homework #1

Math 104: Intro to Analysis

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In this homework, we make a handful of assumptions about the integers. I was verbally told that any assumptions about integers necessary can be made. I have made the following assumptions about integers:

1. The integers are ordered.
2. The integers are closed under addition and multiplication.
3. $2 > 1 > 0$

1 Exercise 1

Let S be the set {rock, paper, scissors}. Define a binary operation $*$: $S \times S \rightarrow S$ via

$$x * y = \begin{cases} \text{the winner of a rock-paper-scissors game between } x \text{ and } y, & \text{if } x \neq y \\ x, & \text{if } x = y. \end{cases}$$

- (a) Prove that $*$ is a commutative operation on S .
- (b) Prove that $*$ is not an associative operation on S .

1.1 Solution (a)

Proof. Commutativity means that $x * y = y * x$.

There are two cases: when $x = y$ and $x \neq y$. Begin with the former. If $x = y$, $x * y = x$ and $y * x = y$. By the transitive property, since $x * y = x$ and $x = y$, $x * y = y$. Also by the transitive property, since $x * y = y$ and $y * x = y$, $x * y = y * x$. Ergo, if $x = y$, $x * y$ is commutative.

Now the case where $x \neq y$. Without loss of generality, suppose that x has the advantage over y (i.e. x is rock when y is scissors, x is scissors when y is paper, etc.). In this case, x will always beat y in a game of ro-sham-bo, so $x * y = x$. Meanwhile, $y * x$ also results in a game of ro-sham-bo, which x has the advantage in and will win, so $y * x = x$. Since $x * y = x$ and $y * x = x$, by the transitive property, $x * y = y * x$.

Ergo, $*$ is a commutative operation on S .

OEΔ

1.2 Solution (b)

Proof by Counterexample. For the associative property to hold, it must be the case that for any r, p , and s in S , $(r * p) * s = r * (p * s)$. Suppose that $r = \text{rock}$, $p = \text{paper}$, and $s = \text{scissors}$. We can perform the internal operations on them. Start with $(r * p) * s$.

$$(r * p) * s = p * s = s = \text{scissors} \tag{1}$$

Then, $r * (p * s)$.

$$r * (p * s) = r * s = r = \text{rock} \tag{2}$$

For associativity to hold, it would require that $s = r$, meaning scissors = rock, which is expressly not true. That acts as a counterexample, so associativity does not hold. OEΔ

2 Exercise 2

Let F be an ordered field with additive identity 0 and multiplicative identity 1.

- (a) Using **only** the field and ordered field axioms, prove that if $a, b \in F$ with $a > 0$ and $0 < b < 1$, then $0 < ab < a$.
- (b) Using **only** the field axioms, prove that if $a, b, c, d \in F$, then $((a+b)+c)+d = d+(c+(b+a)) = ((d+c)+b)+a$.

2.1 Solution (a)

Alvin's Proposition: If $x + y = x + z$, then $y = z$.

Proof. Suppose that (3) is true.

$$x + y = x + z \quad (3)$$

By the additive axiom, there exists some w in F such that $w + x = x + w = 0$. Therefore, we can add w to both sides of (3).

$$w + (x + y) = w + (x + z) \quad (4)$$

Use the associative property of addition to consolidate every $(w + x)$ together.

$$(w + x) + y = (w + x) + z \quad (5)$$

We stated earlier that $w + x = x + w = 0$, so $(w + x) + y = 0 + y$ and $(w + x) + z = 0 + z$. Therefore, $0 + y = 0 + z$. By the definition of the additive identity, $0 + y = y$ and $0 + z = z$. Therefore, $y = z$. OEΔ

Betty's Proposition: $0 \cdot x = 0$.

Proof. Let $x \in F$. Take $0 \cdot x + 0 \cdot x$. By the commutative property, $0 \cdot x + 0 \cdot x = x \cdot 0 + x \cdot 0$. The distributive axiom allows us to combine like terms, so $0 \cdot x + 0 \cdot x = x(0 + 0)$.

The definition of the additive identity is that $a + 0 = a$. If we say that $a = 0$, then $0 + 0 = 0$. Therefore, $0 \cdot x + 0 \cdot x = x \cdot 0$. By the commutative property, $0 \cdot x + 0 \cdot x = 0 \cdot x$. By the definition of the additive identity, $0 \cdot x + 0 \cdot x = 0 \cdot x + 0$.

Note that $0 \cdot x = 0 \cdot x$. Alvin's Proposition tells us that if $x + y = x + z$, then $y = z$. Applying that to $0 \cdot x + 0 \cdot x = 0 \cdot x + 0$, we can say that $0 \cdot x = 0$. OEΔ

Chuck's Proposition: If $a \neq 0$ and $b \neq 0$ then $ab \neq 0$.

Proof by Contradiction. Suppose that $x = 0$ and $y = 0$. Suppose for contradiction that $xy = 0$.

From the definition of the multiplicative inverse, $x \cdot x^{-1} = 1$ and $y \cdot y^{-1} = 1$. Multiply the former by $y \cdot y^{-1}$ on both sides to find that $(x \cdot x^{-1})(y \cdot y^{-1}) = y \cdot y^{-1}$, which transitively tells us that $(x \cdot x^{-1})(y \cdot y^{-1}) = 1$. The following series of equalities are all justified by the associative property (denoted $\stackrel{A}{=}$) and the commutative property (denoted $\stackrel{C}{=}$).

$$1 = (x \cdot x^{-1})(y \cdot y^{-1}) \stackrel{A}{=} ((x \cdot x^{-1}) \cdot y) \cdot y^{-1} \stackrel{C}{=} [x \cdot (x^{-1} \cdot y)] \cdot y^{-1} \quad (6)$$

$$\stackrel{A}{=} [x \cdot (y \cdot x^{-1})] \cdot y^{-1} \stackrel{C}{=} [(x \cdot y) \cdot x^{-1}] \cdot y^{-1} \quad (7)$$

From our initial assumption, we know that $xy = 0$. Therefore, $1 = [0 \cdot x^{-1}] \cdot y^{-1} = 0 \cdot y^{-1} = 0$. This would mean that $0 = 1$, which contradicts that $0 \neq 1$. Ergo, $xy \neq 0$ OEΔ

Derek's Proposition: If $x \neq 0$ and $xy = xz$ then $y = z$.

Proof. Suppose that $x \neq 0$ and $xy = xz$. By the multiplicative axiom, there exists some w in F such that $xw = wx = 1$. Multiply $xy = xz$ by w and get $w(xy) = w(xz)$. The commutative property tells us that $(wx)y = (wx)z$. We can replace each wx with 1 since $wx = 1$ to get $1 \cdot y = 1 \cdot z$. By the definition of the multiplicative identity, $1 \cdot x = x$, so $y = z$. OEΔ

Now for the proof requested: If $a, b \in F$ with $a > 0$ and $0 < b < 1$, then $0 < ab < a$.

Proof. The multiplicative ordered field axiom states that if for some a and b in F , $a \leq b$ and $c \geq 0$, then $ac \leq bc$.

Since $0 < b$ and $0 < a$, $0 \leq b$ and $0 \leq a$. In the same vein, since $b < 1$, $b \leq 1$.

By the multiplicative ordered field axiom, $0 \cdot a \leq ab$ and $ab \leq a$.

By Betty's Proposition, $0 \cdot x = 0$, so $0 \cdot a = 0$ and $0 \geq ab$.

Chuck's Proposition says that if $x \neq 0$ and $y \neq 0$, then $xy \neq 0$. Since $0 < a$, $a \neq 0$. Since $0 < b$, $b \neq 0$. Therefore, $ab \neq 0$. Since $0 \leq ab$ and $0 \neq ab$, $0 < ab$.

Derek's Proposition says that if $x \neq 0$ and $xy = xz$, then $y = z$. Its contrapositive is that if $y \neq z$, then $x = 0$ or $xy \neq xz$. Since $b < 1$, $b \neq 1$. This means that either $a = 0$ or $ab \neq a \cdot 1$. Since $a \neq 0$, it would have to be that $ab \neq a \cdot 1$. By the definition of the multiplicative identity, $a \cdot 1 = a$, so $ab \neq a$.

$ab \leq a$ and $ab \neq a$, so $ab < a$.

Since $0 < ab$ and $ab < a$, $0 < ab < a$. OEΔ

2.2 Solution (b)

Proof. Our first half of the proof will consist of a series of equalities, all of which are justified by the commutative property. The commutative property of addition says that $a + b = b + a$.

$$((a + b) + c) + d = ((b + a) + c) + d \quad (8)$$

$$= (c + (b + a)) + d \quad (9)$$

$$= d + (c + (b + a)) \quad (10)$$

Therefore, $((a + b) + c) + d = d + (c + (b + a))$.

The second half of the proof will consist of a series of equalities, all of which are justified by the associative property. The associative property of addition says that $(a + b) + c = a + (b + c)$, or equivalently that $a + b + c$ is unambiguous.

$$d + (c + (b + a)) = (d + c) + (b + a) \quad (11)$$

$$= ((d + c) + b) + a \quad (12)$$

Therefore, $d + (c + (b + a)) = ((d + c) + b) + a$.

Ergo, transitively, $((a + b) + c) + d = d + (c + (b + a)) = ((d + c) + b) + a$. OEΔ

3 Exercise 3

Let F be a field with order $<$. Recall from class that $(F, <)$ is called an *ordered field* if the following axioms are satisfied:

- (i) For all $a, b, c \in F$ with $a \leq b$, we have $a + c \leq b + c$.
- (ii) For all $a, b, c \in F$ with $a \leq b$ and $0 \leq c$, we have $ac \leq bc$.
- (a) Prove that axiom (ii) is *equivalent* to the axiom
 - (ii)' For all $a, b \in F$ with $a > 0$ and $b > 0$, we have $ab > 0$
 That is, prove that (ii) is true if and only if (ii)' is true. This means that whether we use (ii) or (ii)' in our definition of an ordered field is irrelevant.
- (b) Define a relation $\prec$ on $\mathbb{Q}$ by declaring $a \prec b$ if and only if $b < a$ in the usual order on $\mathbb{Q}$. Prove that $\prec$ is an order on $\mathbb{Q}$ and that axiom (i) is true for $\prec$, but that axiom (ii) fails. Thus $(\mathbb{Q}, \prec)$ is an ordered set but not an ordered field.

Note: The LaTeX commands for $\prec$, $\preceq$, $\succ$, and $\succeq$ are `\prec`, `\preceq`, `\succ`, and `\succeq`, respectively.

3.1 Solution (a)

Proof. First the forward direction, meaning (ii) implies (ii)'.

Take (ii), and suppose we set $a = 0$, rename b to a , and rename c to b . This may be confusing, but what we end up with is the following:

- (ii) For all $a, b \in F$ with $0 \leq a$ and $0 \leq b$, we have $0b \leq ab$.

We proved in class that $0x = 0$. Therefore, $0b = 0$ and we can replace $0b$ with just 0.

- (ii) For all $a, b \in F$ with $0 \leq a$ and $0 \leq b$, we have $0 \leq ab$.

Since this applies for all a and b in F , including those that are not equal to 0, we can set the rule of only considering cases where $a \neq 0$ and $b \neq 0$. This lets us replace $0 \leq a$ and $0 \leq b$ with $0 < a$ and $0 < b$ respectively. Furthermore, we proved in class that if $x \neq 0$ and $y \neq 0$, then $xy \neq 0$. Since $a \neq 0$ and $b \neq 0$, then $ab \neq 0$. Thereby, we can replace all the $\leq$ signs with $<$ signs.

- (ii) For all $a, b \in F$ with $a > 0$ and $b > 0$, we have $ab > 0$.

This gets us to (ii)'.

Now for the backwards direction.

Since $a > 0$ and $b > 0$, without loss of generality, $0 < a \leq b$. Focus on the latter two terms, $a \leq b$. The ordered field axiom (i) tells us we can add $-a$, the additive inverse of a , to all sides and end up with $a - a \leq b - a$. By the additive inverse axiom, we know that $a - a = 0$. Therefore, $0 \leq b - a$. Let there also be some $0 \leq c$. From here, we bifurcate our proof between (1) $0 < b - a$ and $0 < c$, (2) $0 < b - a$ and $0 = c$, (3) $0 = b - a$ and $0 < c$, and (4) $0 = b - a$ and $0 = c$ cases.

Case 1: $0 < b - a$ and $0 < c$. By ordered field axiom (ii)', since $0 < b - a$ and $0 < c$, $0 < c(b - a)$. By the distributive property, $c(b - a) = bc - ac$, so $0 < bc - ac$. The additive inverse axiom tells us that there exists some $-(-ac)$ such that $ac + (-(-ac)) = 0$. Ordered field axiom (i) tells us we can add $-(-ac)$ to all sides, so $-(-ac) < bc - ac - (-ac)$. In class, we proved that

$-(-x) = x$, so $-(-ac) = ac$, which we can replace in the equation to get $ac < bc - ac + ac$. The commutative property lets us say that $ac < bc + ac - ac$. By the additive inverse axiom, $ac - ac = 0$. Therefore, $ac < bc + 0$. By the additive identity, $bc + 0 = bc$, so $ac < bc$, and by extension $ac \leq bc$.

Case 2: $0 < b - a$ and $0 = c$. If we multiply $(b - a)$ by c , it is the equivalent to multiplying $(b - a)$ by 0. This means that $c(b - a) = 0(b - a)$. In class, we proved that for all x , $0x = 0$. This means that $c(b - a) = 0(b - a) = 0$. The distributive property says that $c(b - a) = bc - ac$. Putting this together, we find $bc - ac = 0$. We can add $-(-ac)$, rewritten as ac , to both sides to find $bc - ac + ac = ac$. By the commutative property, $bc + ac - ac = ac$. By the additive identity axiom, $bc = ac$, and by extension $ac \leq bc$.

Case 3: $0 = b - a$ and $0 < c$. If we multiply $(b - a)$ by c , it is the equivalent to multiplying c by 0. This means that $c(b - a) = c \cdot 0$. By the commutative property of multiplication, $c(b - a) = (b - a)c$ and $c \cdot 0 = 0c$, so $(b - a)c = 0c$. In class, we proved that for all x , $0x = 0$. This means that $(b - a)c = 0c = 0$. The distributive property says that $(b - a)c = bc - ac$. Putting this together, we find $bc - ac = 0$. We can add $-(-ac)$, which we proved in class is equal to and can be rewritten as ac , to both sides to find $bc - ac + ac = ac$. By the commutative property, $bc + ac - ac = ac$. By the additive inverse axiom, $ac - ac = 0$, so $bc + 0 = ac$. By the additive identity axiom, $bc + 0 = bc$, so $bc = ac$, and by extension $ac \leq bc$.

Case 4: $0 = b - a$ and $0 = c$. We can add $-(-a)$, which we proved in class is equal to a , to $0 = b - a$ to get $a = b - a + a$. The commutative property tells us that $a = b + a - a$. The additive inverse axiom says that $a = b + 0$. The additive identity axiom says that $b = b + 0$, so $a = b$. Multiply both sides by c to get $ac = bc$, and by extension $ac \leq bc$.

With all cases satisfied, this proves that if $a \leq b$ and $c \geq 0$, $ac \leq bc$.

OEΔ

3.2 Solution (b)

Proof. The textbook says that $\mathbb{Q}$ is an ordered set if $r > s$ is defined to mean that $r - s$ is a positive rational number. The textbook also says (on page 7) that the rational numbers are an ordered field.

We start by proving that $\prec$ is an order on $\mathbb{Q}$. We know that either $a < b$, $a = b$, or $b < a$. $a < b$ means $a \succ b$. $b < a$ means $b \succ a$. This tells us that either $a \succ b$, $a = b$, or $b \succ a$. Next, suppose that $a \succ b$ and $b \succ c$. This means that $a < b$ and $b < c$. We know from the first ordered field axiom that if $a < b$ and $b < c$, then $a < c$, so in this case $a < c$. Because of that, we can say that $a \succ c$. In total, if $a \succ b$ and $b \succ c$, then $a \succ c$. This all means that $\prec$ is an order on $\mathbb{Q}$.

Next, we prove that for all a, b , and c in $\mathbb{Q}$ with $a \leq b$, we have $a + c \leq b + c$. That is to say, for all a, b , and c in $\mathbb{Q}$ with $a \geq b$, we have $a + c \geq b + c$. Take the first field axiom, applied to the rational numbers.

- (i) For all $a, b, c \in \mathbb{Q}$ with $a \leq b$, we have $a + c \leq b + c$.

Swap a and b , without loss of generality.

- (i) For all $a, b, c \in \mathbb{Q}$ with $b \leq a$, we have $b + c \leq a + c$.

If $b < a$, then $b \succ a$. Therefore, if $b < a$ or $b = a$, then $b \succ a$ or $b = a$. Therefore, if $b \leq a$, then $b \succeq a$. Therefore, we can swap every $\leq$ with $\succeq$.

- (i) For all $a, b, c \in \mathbb{Q}$ with $b \succeq a$, we have $b + c \succeq a + c$.

Ergo, axiom (i) is true for $\prec$.

Axiom (ii) can be proven impossible by counterexample. First, we restate the axiom for $<$.

(ii) For all $a, b, c \in F$ with $a \leq b$ and $0 \leq c$, we have $ac \leq bc$.

Replace $<$ with $\prec$ and $\leq$ with $\preceq$.

(ii) For all $a, b, c \in F$ with $a \preceq b$ and $0 \preceq c$, we have $ac \preceq bc$.

Based on the definition of $\prec$, we can replace $a \preceq b$ with $a \geq b$, $0 \preceq c$ with $0 \geq c$, and $ac \preceq bc$ with $ac \geq bc$.

(ii) For all $a, b, c \in F$ with $a \geq b$ and $0 \geq c$, we have $ac \geq bc$.

Suppose that $a = 2$, $b = 1$, and $c = -1$. Since $2 \geq 1$, $a \geq b$. Since $0 \geq -1$, $0 \geq c$. ac is equal to -2 and bc is equal to -1 . Since $-2 < -1$, $ac < bc$, which means that $ac \not\geq bc$, which contradicts the second field axiom. As such, axiom (ii) fails.

Ergo, $(\mathbb{Q}, \prec)$ is an ordered set but not an ordered field.

OE Δ

4 Exercise 4

Let $S \subset \mathbb{R}$ be nonempty and bounded from above. Let $a = \sup S$.

- (a) Show that, for any real $\epsilon > 0$, there is $x \in S$ with $x > a - \epsilon$.
- (b) Suppose $a \notin S$. Show that, for any real $\epsilon > 0$, there are infinitely many elements of S in the open interval $(a - \epsilon, a)$.

4.1 Solution (a)

Proof by Contradiction. Suppose for contradiction that, for any real $\epsilon > 0$, there does not exist $x \in S$ with $x > a - \epsilon$. This means that, for all $x \in S$, $x \leq a - \epsilon$. This means that $a - \epsilon$ is an upper bound of S .

We know that $\epsilon > 0$. Add a to both sides of S , which through the first ordered field axiom and the fact that $\mathbb{R}$ is an ordered field means that $a + \epsilon > a$. By the additive inverse property of fields ($\mathbb{R}$ included), there exists some $-\epsilon$ such that $\epsilon + (-\epsilon) = 0$. Add $-\epsilon$ to both sides of $a + \epsilon > a$ to get $a + \epsilon - \epsilon > a - \epsilon$. By the additive inverse property of fields, $a + \epsilon - \epsilon = a > a - \epsilon$.

Therefore, there exists an upper bound of S that is less than a in the form of $a - \epsilon$. Therefore, a is not the least upper bound of S . This contradicts that a is the least upper bound of S , so our initial assumption must be false. Ergo, for any real $\epsilon > 0$, there is $x \in S$ with $x > a - \epsilon$. $\text{OE}\Delta$

4.2 Solution (b)

Proof. An infinite number of elements in the interval means that the elements can be mapped onto the natural numbers or a set with a larger cardinality.

We can define a set E recursively. Suppose that the first element of E , denoted e_0 , is equal to ϵ . As established in part (a), there exists some x in S with $x > a - \epsilon$. We can create a new set X and call the x in S with $x > a - \epsilon$, x_1 . Since a is the least upper bound of S , it is an upper bound of S , meaning that because x_1 is in S , $x_1 < a$.

Now, suppose that we have element n (in the natural numbers) of x , denoted x_n , which is an element of S . Since x_n is an element of S and a is an upper bound of S , $x_n < a$. Add the additive inverse of x_n , $-x_n$, to both sides, which the first ordered field axiom lets us do, and we get $x_n - x_n < a - x_n$. By the additive inverse definition, $x_n - x_n = 0$, so $0 < a - x_n$. Since $\mathbb{R}$ is a field and fields are closed under addition, $a - x_n$ is a real number. We can denote $a - x_n$ as e_n . Since e_n is real and greater than 0, it is possible that some new $\epsilon = e_n$. Now take $a - e_n$. As established in part (a), there exists some x in S with $x > a - e_n$. We can call the x in S with $x > a - e_n$ that we just discovered existed x_{n+1} . Therefore, $x_{n+1} > a - e_n$. Substitute in the definition of $e_n = a - x_n$ to get $x_{n+1} > a - (a - x_n)$. By the commutative property, $a - (a - x_n) = a - a + x_n = (a - a) + x_n$. By the additive inverse axiom, $a - a + x_n = 0 + x_n$. By the additive identity axiom, $0 + x_n = x_n$. Putting everything together transitively, $x_{n+1} > x_n$. This guarantees that every x_{n+1} is distinct from x_n . Transitively, this inequality also proves that x_{n+1} is distinct from every other element of X . Furthermore, since a is an upper bound of S , for every x_n , $x_n < a$.

This pairs up our subset of the set of elements of S in the open interval $(a - \epsilon, a)$ with the natural numbers. This means that the cardinality of the natural numbers is less than or equal to the cardinality of the set of elements of S in the open interval $(a - \epsilon, a)$. The cardinality of the natural numbers is countably infinite. Ergo, the cardinality of the set of elements of S in the open interval $(a - \epsilon, a)$ is infinite. $\text{OE}\Delta$

5 Exercise 5

For an integer $m \geq 2$, we define $\mathbb{Z}/m\mathbb{Z}$ to be the set $\{0, 1, \dots, m-1\}$, together with the operations of addition and multiplication modulo m . That is, to add two elements of m , we add them as we normally would and then take the remainder upon division by m . Multiplication is defined similarly. Thus, in $\mathbb{Z}/5\mathbb{Z}$ for example, we have $2 + 4 = 3$ and $4 \cdot 4 = 1$.

You may assume that all of the axioms for addition, multiplication, and distributivity from fields also hold in $\mathbb{Z}/m\mathbb{Z}$ (with 0 and 1 as the identities), *except* for the axiom

- (*) every nonzero element has a multiplicative inverse
which is the subject of this problem.

- (a) Show that if m is a composite number, then axiom (*) fails.
(b) Show that if m is a prime number, then axiom (*) holds. That is, $\mathbb{Z}/p\mathbb{Z}$ is a field with only finitely many elements!

Hint for part (b): You may use *Bézout's identity* from number theory: if $a, b \in \mathbb{Z}$ are coprime then there are $x, y \in \mathbb{Z}$ so that $ax + by = 1$. Then use that for a prime p , the integers $1, 2, \dots, p-1$ are all coprime to p . You may also use the *Euclidean division theorem*: namely if b is a positive integer, then any $a \in \mathbb{Z}$ may be written uniquely as $a = bq + r$, where $0 \leq r < b$.

5.1 Solution (a)

Within this and the next proof, there is a proposition about equalities in $\mathbb{Z}/m\mathbb{Z}$ that should be made.

Gabby's Proposition: For some a in $\mathbb{Z}/m\mathbb{Z}$ and any integer n , $nm + a = a$ within $\mathbb{Z}/m\mathbb{Z}$.

Proof. Take nm . By the Euclidean Division Theorem, if b is a positive integer, then any a in $\mathbb{Z}$ may be written uniquely as $a = bq + r$ for $0 \leq r < b$. This would therefore apply to nm for $b = m$, so $nm = nm + r$. By the definition of the additive identity, $nm = nm + 0$. Transisistively, this means $nm + 0 = nm + r$. By Alvin's Proposition (see Exercise 2, also proven in class), that means $r = 0$, so the remainder of nm upon division by m is 0. That means that in $\mathbb{Z}/m\mathbb{Z}$, $nm = 0$. Add a to both sides to get that $nm + a = a$ in $\mathbb{Z}/m\mathbb{Z}$. OEΔ

Now for the proof that if m is a composite number, then axiom (*) fails.

Proof. Having a multiplicative inverse in a set means that for any number a in that set there exists some number a^{-1} also in that set for which $a \cdot a^{-1} = 1$. If m is a composite number, that means that there exists some two other integers j and k greater than or equal to 2 such that $m = j \cdot k$.

Without loss of generality, focus on j . We know that $j \geq 2$. Since $2 > 1$, by the transitive property (if $j > 2$, substitution if $j = 2$), $j > 1$. If we add this to, for any $n \geq 0$, $njk = njk$ (which comes from the reflexive property), we get $njk + j > njk + 1$. By the distributive property, $j(nk + 1) > njk + 1$. We also know that $1 > 0$, which can be added to $njk = njk$ to get $njk + 1 > njk + 0$. By the additive identity axiom, $njk + 0 = njk$, so $njk + 1 > njk$. Putting these together, we get $j(nk + 1) > njk + 1 > njk$.

This tells us that $njk + 1$ will always be between njk and $njk + j$, which are consecutive multiples of j , but will not be equal to either of them so long as $j \geq 2$. Taking the modulus of this past statement of base jk , we wind up with $j > 1 > 0$, showing that in $\mathbb{Z}/m\mathbb{Z}$, 1 will always be between 0 and j , which are consecutive multiples of j , but will not be equal to either of them so long as $j \geq 2$. Ergo, in $\mathbb{Z}/m\mathbb{Z}$ where m is composite, the axiom (*) will fail. OEΔ

5.2 Solution (b)

Proof. For the multiplicative identity to hold in set $\mathbb{Z}/p\mathbb{Z}$ for prime p , for all integers b with $0 < b < p$, there must exist some integers x and y with $0 < y < p$ such that, in the set of all integers, $by = px + 1$.

Recall Bézout's identity, which says that if integers a and b are coprime, then there exist integers x and y such that $ax + by = 1$. In our set $\mathbb{Z}/p\mathbb{Z}$, p is prime, so it is coprime with every number in the set $\{1, 2, \dots, p-1\}$. Set a to be equal to p , and suppose b is in the set $\{1, 2, \dots, p-1\}$. This means that there exist integers x and y such that $px + by = 1$. Without loss of generality and for the sake of cleanliness, if we replace every instance of x with $-x$ and use the commutative property, we get $by - px = 1$. Adding px (the additive inverse of $-px$) to both sides, we get $by - px + px = 1 + px$. The commutative property tells us that $by + px - px = px + 1$. The definition of the additive inverse tells us that $by + 0 = px + 1$. The additive identity tells us that $by = px + 1$. By Gabby's Proposition, $px + 1 = 1$ within $\mathbb{Z}/m\mathbb{Z}$. Transistively, this means that $by = 1$ within $\mathbb{Z}/m\mathbb{Z}$. If we can ensure that this would hold in a case where $0 < y < p$, then we can prove the axiom holds.

Since there is the requirement of $b < p$, we can multiply both sides by y to find that $by < py$. We want to find some $y < p$. We can multiply both sides of the inequality by positive nonzero integer b to find that we want to find some y such that $by < bp$. If $y < p$ already, we're in good company and the rest of this proof is moot. If not, then $y \geq p$, which we can multiply by b (which is a positive integer) to find that we have $by \geq bp$. By the Euclidean division theorem, because both by and bp are positive integers, by can be written uniquely as $by = bpq + r$ for some integer q where $r < bp$. By the additive inverse, $by - bpq = r$. Since $r \geq 0$, $by - bpq \geq 0$. By the distributive property, $b(y - pq) = r$, so there is an integer multiple of b that is equal to r . Since $b(y - pq) = r$ and $r < bp$, then $b(y - pq) < bp$. bpq is a multiple of p and bq , with bq being an integer because integers are closed under multiplication. By the same argument, $-bpq$ is an integer multiple of p . We have established that in $\mathbb{Z}/p\mathbb{Z}$, $by = 1$. Therefore, by Gabby's proposition, we can safely say that in $\mathbb{Z}/p\mathbb{Z}$, $by - bpq = 1$. By the distributive property, $b(y - pq) = 1$ in $\mathbb{Z}/p\mathbb{Z}$. That means that if we set $y' = y - pq$, then $by' < bp$, by extension $y' < p$, and $by' = 1$ in $\mathbb{Z}/p\mathbb{Z}$. Ergo, axiom (*) holds. OEΔ

6 Exercise 6

Let F be a finite field with identities 0 and 1.

- (a) Show that there is an integer $n \geq 2$ so that $\underbrace{1 + 1 + \cdots + 1}_{n \text{ times}} = 0$. (*Hint:* by the pigeonhole principle, at least two different sums of the form $1 + 1 + \cdots + 1$ must be equal)
- (b) By using the fact that $\underbrace{1 + 1 + \cdots + 1}_{ab \text{ times}} = (\underbrace{1 + 1 + \cdots + 1}_{a \text{ times}})(\underbrace{1 + 1 + \cdots + 1}_{b \text{ times}})$ for $a, b \geq 2$, show that the *smallest* possible integer n from part (a) is prime.
- (c) Using part (a), show that there are no finite ordered fields.

6.1 Solution (a)

Irwin's Proposition: Any sum of the form $\underbrace{1 + \cdots + 1}_{n \text{ times}}$ is in F .

Proof by Induction. The base case is where $n = 1$ and the sum is just 1, which we know is in F .

The inductive step comes first from $\underbrace{1 + \cdots + 1}_{n \text{ times}}$, which we assume is in F . Add 1 (in F) to this and we get $\underbrace{1 + \cdots + 1}_{n \text{ times}} + 1$, which is in F because F is a field, which is closed under addition. $\underbrace{1 + \cdots + 1}_{n \text{ times}} + 1$ can be rewritten as $\underbrace{1 + \cdots + 1}_{n+1 \text{ times}}$. Ergo, $\underbrace{1 + \cdots + 1}_{n \text{ times}}$ being in F implies $\underbrace{1 + \cdots + 1}_{n+1 \text{ times}}$ being in F .

Therefore, any sum $\underbrace{1 + \cdots + 1}_{n \text{ times}}$ is in F .

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Now we prove that there is an integer $n \geq 2$ so that $\underbrace{1 + 1 + \cdots + 1}_{n \text{ times}} = 0$.

Proof. The key term here is finite. This means that the cardinality of F is some finite number k . F being a field, a field axiom says that it is closed under addition. We know that 1, being the multiplicative identity, is contained in F . The additive identity 0 is also contained in F .

Irwin's Proposition says that any sum of the form $\underbrace{1 + \cdots + 1}_{n \text{ times}}$ is in F . F is finite, and therefore by the Pigeonhole Principle, there must be two sums of the forms $\underbrace{1 + \cdots + 1}_{m \text{ times}}$ and $\underbrace{1 + \cdots + 1}_{n \text{ times}}$ such that $m \neq n$ where $\underbrace{1 + \cdots + 1}_{m \text{ times}} = \underbrace{1 + \cdots + 1}_{n \text{ times}}$. Without loss of generality, suppose that $m > n$. This is possible because m and n are integers, which are ordered. By the additive inverse axiom, there exists some $-\left(\underbrace{1 + \cdots + 1}_{n \text{ times}}\right)$ such that $\underbrace{1 + \cdots + 1}_{n \text{ times}} - \left(\underbrace{1 + \cdots + 1}_{n \text{ times}}\right) = 0$. We can add this inverse to the earlier inequality to find that $\underbrace{1 + \cdots + 1}_{m \text{ times}} - \left(\underbrace{1 + \cdots + 1}_{n \text{ times}}\right) = \underbrace{1 + \cdots + 1}_{n \text{ times}} - \left(\underbrace{1 + \cdots + 1}_{n \text{ times}}\right)$. By the

earlier additive inverse, $\underbrace{1 + \cdots + 1}_{m \text{ times}} - \left(\underbrace{1 + \cdots + 1}_{n \text{ times}} \right) = 0$. We can distribute the negatives in the second term to get $\underbrace{1 + \cdots + 1}_{m \text{ times}} + \left(\underbrace{-1 - \cdots - 1}_{n \text{ times}} \right) = 0$. Since $m > n$, we can pair up every number from the latter term with one from the former term, ending up with $\underbrace{(1 - 1) + \cdots + (1 - 1)}_{n \text{ times}} + \underbrace{1 + \cdots + 1}_{m-n \text{ times}} = 0$. By the additive inverse, $1 - 1 = 0$, so $\underbrace{0 + \cdots + 0}_{n \text{ times}} + \underbrace{1 + \cdots + 1}_{m-n \text{ times}} = 0$. The additive identity tells us that $a + 0 = 0 + a = a$, so we can say that $\underbrace{1 + \cdots + 1}_{m-n \text{ times}} = 0$.

Since $m > n$, $m - n > 0$. Integers being closed under addition, $m - n$ would have to be an integer. Therefore, since $m - n > 0$, the minimum possible integer it could be is 1, so $m - n \geq 1$. Now, suppose that $m - n = 1$. That would mean that $0 = \underbrace{1 + \cdots + 1}_{m-n \text{ times}} = \underbrace{1 + \cdots + 1}_{1 \text{ time}} = 1$. This would mean that $1 = 0$, which is impossible in a field, so $m - n \neq 1$. The next smallest integer that $m - n$ could be is 2, so $m - n \geq 2$. Ergo, for integer $p = m - n \geq 2$, $\underbrace{1 + \cdots + 1}_{p \text{ times}} = 0$. OEΔ

6.2 Solution (b)

Proof by Contradiction. Suppose for contradiction that n is composite. Suppose also that n is the smallest integer such that $\underbrace{1 + \cdots + 1}_{n \text{ times}} = 0$. There exists some two integers a and b such that $n = ab$ and (without loss of generality between a and b) $1 < a \leq b < n$. If n is prime, then one of a or b is equal to 1, and the other is equal to n .

Take the case of $\underbrace{1 + \cdots + 1}_{n \text{ times}}$. By the fact we are given and our earlier statement that $n = ab$, we can say that $\underbrace{1 + \cdots + 1}_{n \text{ times}} = \underbrace{1 + \cdots + 1}_{ab \text{ times}} = \underbrace{(\underbrace{1 + \cdots + 1}_{a \text{ times}}) (\underbrace{1 + \cdots + 1}_{b \text{ times}})}_{ab \text{ times}}$. We can say that $\underbrace{1 + \cdots + 1}_{a \text{ times}} = \alpha$ and $\underbrace{1 + \cdots + 1}_{b \text{ times}} = \beta$. Therefore, $\underbrace{(\underbrace{1 + \cdots + 1}_{a \text{ times}}) (\underbrace{1 + \cdots + 1}_{b \text{ times}})}_{ab \text{ times}} = \alpha\beta$.

In class, we proved that if $x \neq 0$ and $y \neq 0$, then $xy \neq 0$. The contrapositive of this says that if $xy = 0$, then $x = 0$ or $y = 0$. Since $\alpha\beta = 0$, then $\alpha = 0$ or $\beta = 0$. This means that either $\underbrace{1 + \cdots + 1}_{a \text{ times}} = 0$ or $\underbrace{1 + \cdots + 1}_{b \text{ times}} = 0$.

According to the textbook, the rationals are an ordered field (p.7). That means that if $x > 0$ and $y < z$, then $xy < xz$ (p.8). We know that $b \geq 2$, meaning $b > 1$. We also know that $a \geq 2$ meaning $a > 0$. Therefore, $a < ab$. We can use the same logic with swapped variables to show that $b < ab$. We earlier established that either $\underbrace{1 + \cdots + 1}_{a \text{ times}} = 0$ or $\underbrace{1 + \cdots + 1}_{b \text{ times}} = 0$. Since both a and b are less than ab and $n = ab$, that means that there is some number c (which is either a or b) less than n such that $\underbrace{1 + \cdots + 1}_{c \text{ times}} = 0$. This contradicts that n is the *smallest* integer for which the property holds. Therefore, n cannot be composite. Ergo, n is prime.

The reason that this does not work when n is prime is that the only integer divisors of n are 1 and n . It cannot be that $\underbrace{1 + \cdots + 1}_{1 \text{ time}} = 0$ because $\underbrace{1 + \cdots + 1}_{1 \text{ time}} = 1$, which would mean that $1 = 0$,

cannot be true. If $\underbrace{1 + \cdots + 1}_{n \text{ times}} = 0$, that is just circular logic, so it's already true and gives us no new information. OEΔ

6.3 Solution (c)

Proof by Contradiction. Suppose for contradiction that F is an ordered field. Since an ordered field is an ordered set which satisfies a handful of axioms, that means F is an ordered set.

In part (a), we proved that in a finite field, there exists some number $n \geq 2$ such that $\underbrace{1 + \cdots + 1}_{n \text{ times}} = 0$. Part (b) established that there is a smallest integer n where that is the case.

Associatively, $\underbrace{1 + \cdots + 1}_{n \text{ times}}$ is equal to $\underbrace{1 + \cdots + 1}_{n-1 \text{ times}} + 1$.

One of the axioms of the ordered fields is that if x, y , and z are in F and $y < z$, then $x + y < x + z$. We can prove by induction that $0 < \underbrace{1 + \cdots + 1}_{n \text{ times}}$. We know that $0 < 1$, which acts as our base case.

For the inductive step, we assume that $0 < \underbrace{1 + \cdots + 1}_{k \text{ times}}$ and add $\underbrace{1 + \cdots + 1}_{n \text{ times}}$ to both sides of $0 < 1$. We end up with $0 + \underbrace{1 + \cdots + 1}_{k \text{ times}} < 1 + \underbrace{1 + \cdots + 1}_{k \text{ times}}$. By the commutative property and the additive identity

being 0, $\underbrace{1 + \cdots + 1}_{k \text{ times}} < \underbrace{1 + \cdots + 1}_{k \text{ times}} + 1$. We previously established that $\underbrace{1 + \cdots + 1}_{k \text{ times}} + 1 = \underbrace{1 + \cdots + 1}_{k+1 \text{ times}}$.

Transistively, this means $0 < \underbrace{1 + \cdots + 1}_{k+1 \text{ times}}$. Therefore, for all natural numbers k (which integers $k \geq 2$ is a subset of), $0 < \underbrace{1 + \cdots + 1}_{k \text{ times}}$.

Suppose now that for some $n \geq 2$, $0 = \underbrace{1 + \cdots + 1}_{n \text{ times}}$, which we established as possible in part

(a). According to our inductive proof, for that $n \geq 2$, $0 < \underbrace{1 + \cdots + 1}_{n \text{ times}}$. Both of these being true

contradicts the first axiom of the ordered sets, which says that one and only one of the statements in the set $\{x < y, x = y, y < x\}$ is true. Therefore, our initial assumption is false. Ergo, F is not an ordered field and no finite ordered field exists. OEΔ